Security Incident Investigation Using Splunk and pfSense: Infection Monkey Simulation

1. Executive Summary:

This project focuses on investigating a simulated cyberattack in a virtual network using Infection Monkey, pfSense, and Splunk. The lab environment included a pfSense firewall, an internal workstation, and three Linux machines in the DMZ network. Infection Monkey was used to simulate attacks and identify security weaknesses in the network.

During the simulation, Infection Monkey discovered seven open services across four machines and successfully compromised one machine, TargetLinux02 (172.31.0.30), through a VSFTPD backdoor vulnerability. I used Splunk to examine firewall logs, investigate suspicious network activity, and look for evidence of the attack. The goal of this project was to understand how a SOC analyst investigates security incidents, evaluates security alerts, and recommends improvements to strengthen network security.

2. Lab Environment:

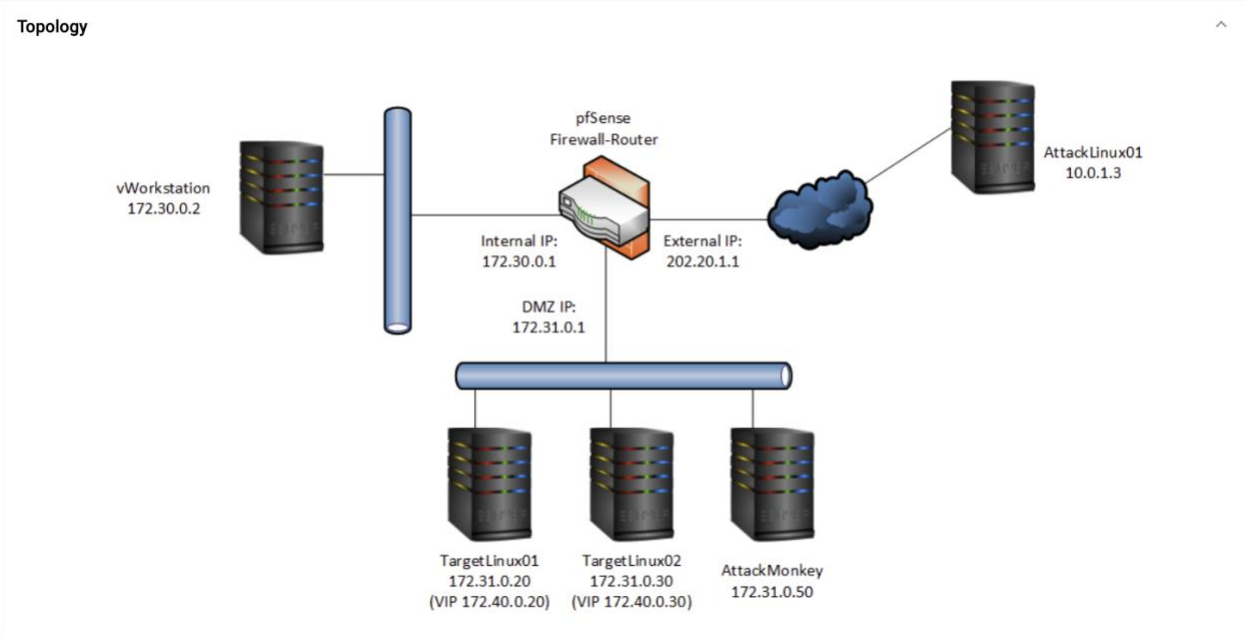


Figure 1. Virtual lab network topology showing the pfSense firewall, internal workstation, DMZ, and attack simulation systems.

This project was conducted in a virtual lab environment consisting of a pfSense firewall-router, an internal workstation (vWorkstation), and three Linux machines in the DMZ network. The vWorkstation (172.30.0.2) was used to access the pfSense web interface and run Splunk for collecting and analyzing firewall logs.

The pfSense firewall was configured to manage network traffic between the internal LAN, DMZ, and external network. The DMZ included TargetLinux01 (172.31.0.20), TargetLinux02 (172.31.0.30), and AttackMonkey (172.31.0.50). Infection Monkey was used to simulate attacks against systems in the lab, while Splunk was used to investigate network activity and identify suspicious events.

Tools Used:

- pfSense: Firewall configuration, network segmentation, and traffic logging.
- Splunk: Security log collection, searching, and investigation.
- Infection Monkey: Simulated attack activity and security assessment.
- Linux: Target systems used in the simulated network environment.

3. Investigation Objectives:

The purpose of this investigation was to analyze suspicious network activity generated by the Infection Monkey simulation and determine what security events could be identified using Splunk and pfSense firewall logs.

The main objectives were to:

1. Identify suspicious network connections between the attacking and targeted machines.
2. Analyze firewall logs in Splunk to determine which IP addresses and services were involved.
3. Investigate whether the available logs showed evidence of the simulated attack.
4. Compare the Splunk findings with the Infection Monkey report to understand the limitations of the available security logs.
5. Identify potential security weaknesses and recommend improvements to network monitoring and protection.

4. Detection and Analysis:

4.1 Log Collection and Verification:

I began my investigation by verifying that Splunk was receiving and storing security logs from the pfSense firewall. I searched the pfsense_syslog index and examined the raw events to understand what information was available for investigation.

The first search displayed firewall logs containing timestamps, IP addresses, network protocols, and firewall actions. Several events showed blocked ICMP traffic from 202.20.1.2 to 202.20.1.1.

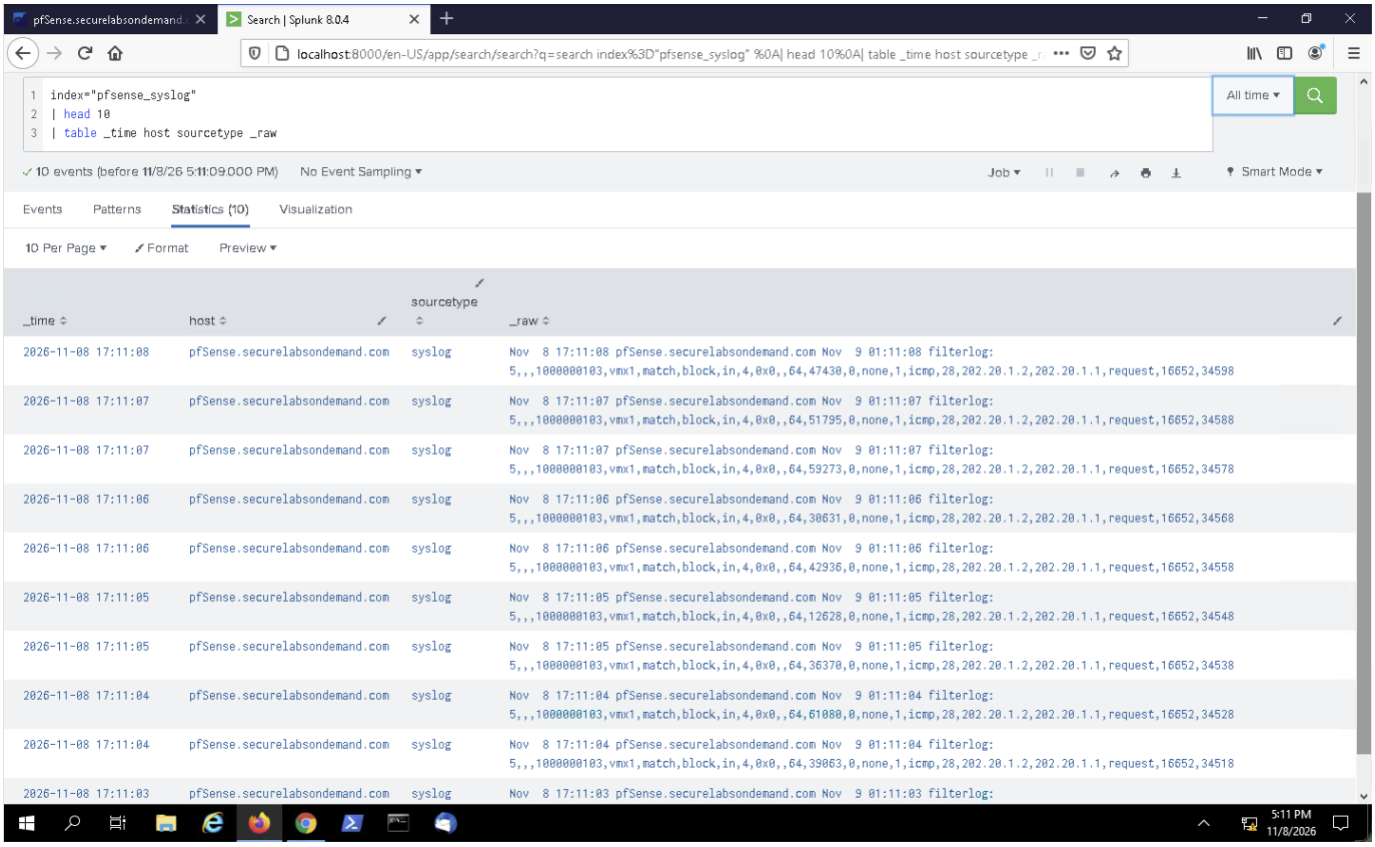


Figure 2. Raw pfSense firewall logs showing blocked ICMP traffic in Splunk.

I then used the stats command to summarize the collected events. The results showed 8,084 events from pfSense.securelabsondemand.com, with timestamps ranging from November 8, 2026, at 4:59:10 PM to 5:17:56 PM. All events were identified with the syslog sourcetype.

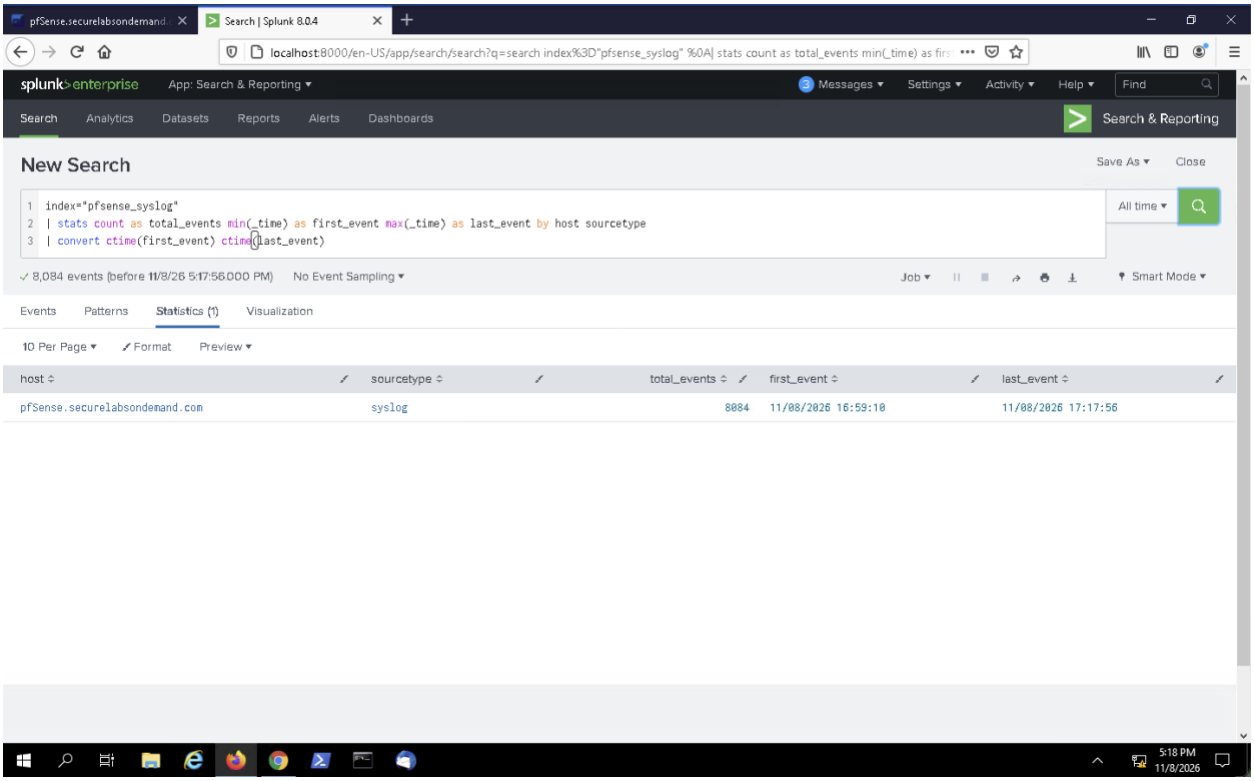


Figure 3. Splunk statistics showing the total number of collected events and their timestamp range.

These results confirmed that Splunk contained security logs that could be used to monitor network activity and investigate suspicious events.

4.2 Initial Threat Detection and Suspicious IP Investigation:

After verifying the collected security logs, I began investigating suspicious activity without identifying a specific attacking machine in advance. I searched the *pfsense_syslog* index for Snort intrusion detection alerts to identify potential security threats.

The search returned 338 events, including several high-priority alerts associated with possible exploitation attempts. One notable alert was "ET WEB_SERVER Possible CVE-2014-6271 Attempt in Headers," which indicated possible attempts to exploit the Shellshock vulnerability.

The alerts showed suspicious traffic originating from 172.31.0.50 and targeting 172.31.0.20 and 172.31.0.30 on TCP port 80.

Based on these findings, I identified 172.31.0.50 as an IP address requiring further investigation because it appeared in multiple high-priority security alerts targeting different systems. However, the Snort alerts alone did not confirm successful exploitation.

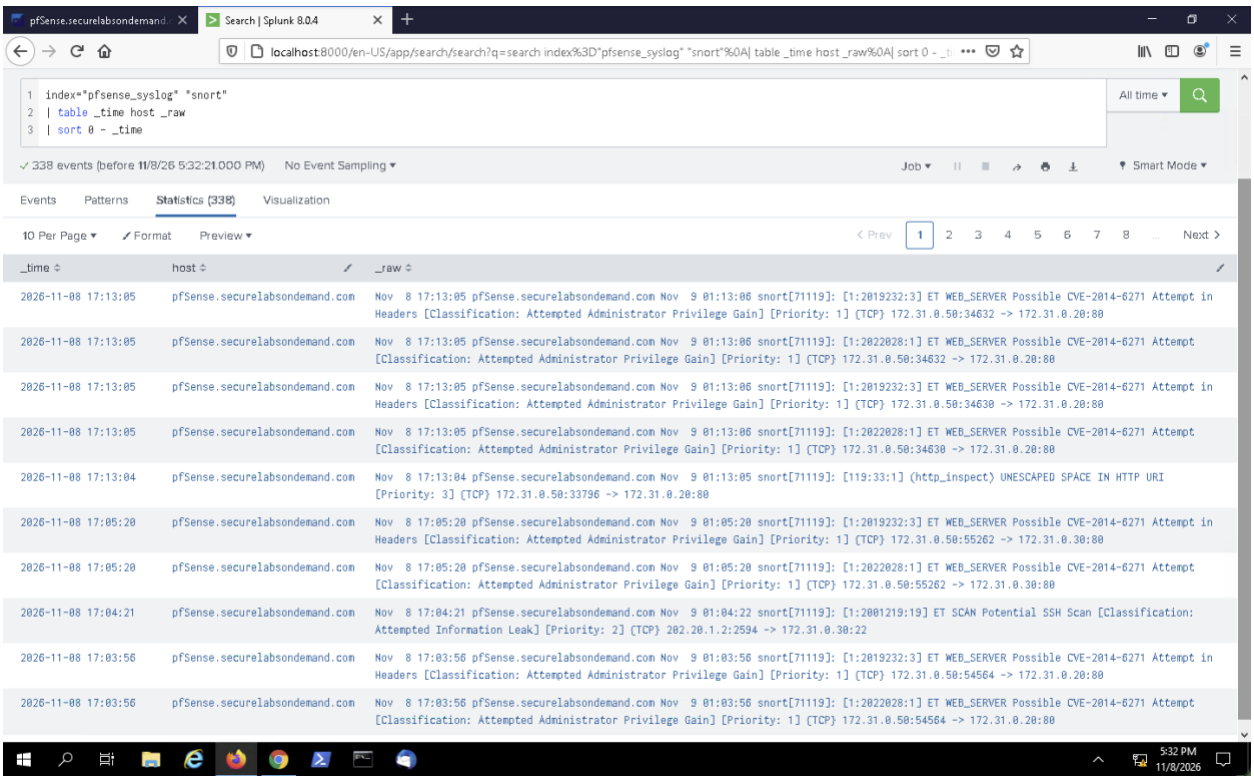


Figure 4. Splunk search results showing Snort intrusion detection alerts, including possible Shellshock exploitation attempts and SSH scanning activity.

4.3 Event Correlation and Timeline:

After identifying 172.31.0.50 as a suspicious source IP, I used Splunk to search for Snort alerts involving this address and sorted the results chronologically. The search returned 12 events showing suspicious activity targeting two Linux machines.

The earliest visible alert occurred on November 8, 2026, at 5:03:54 PM, when Snort detected an unusual HTTP request targeting 172.31.0.20 on TCP port 80. Two seconds later, Snort generated alerts indicating possible Shellshock exploitation attempts against the same machine.

At 5:05:20 PM, additional Shellshock-related alerts appeared involving traffic from 172.31.0.50 to 172.31.0.30. Later, at approximately 5:13 PM, Snort detected more suspicious HTTP activity targeting 172.31.0.20.

By comparing the timestamps, source and destination IP addresses, and alert descriptions, I identified a pattern of repeated suspicious activity involving multiple machines. However, the alerts alone did not confirm successful exploitation.

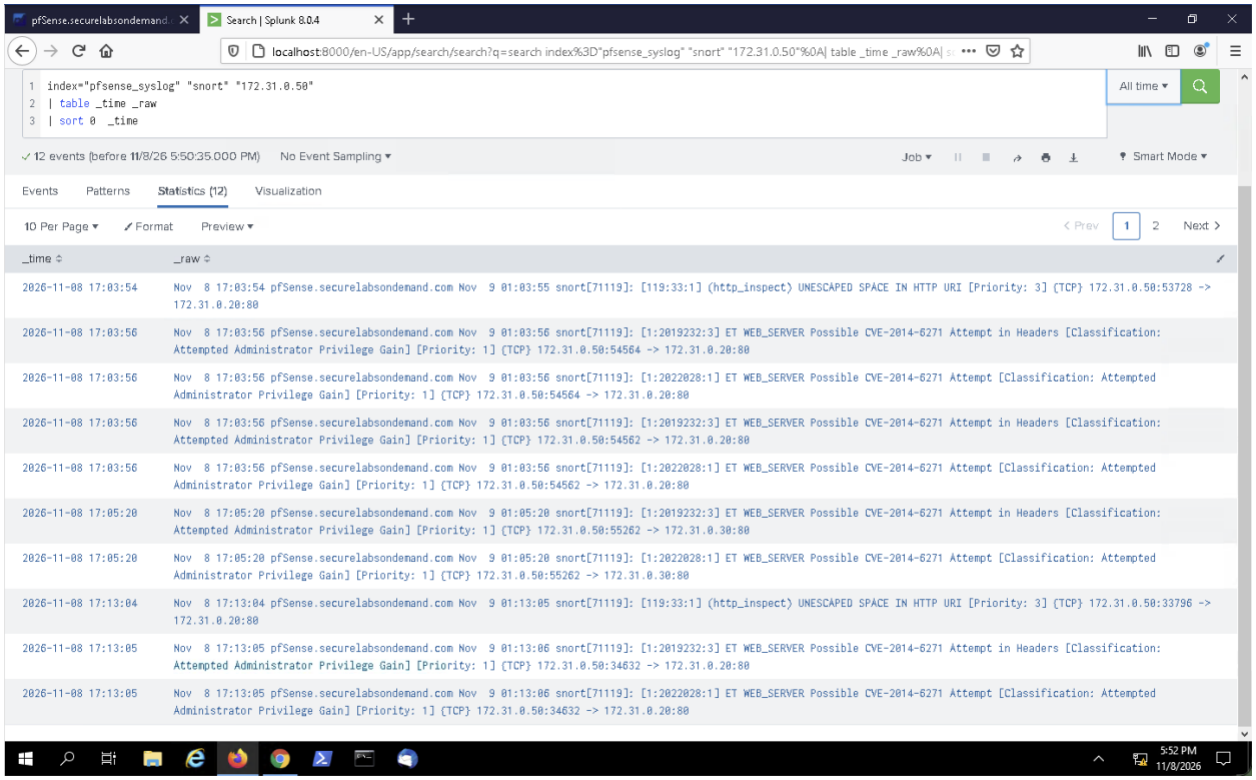


Figure 5. Splunk search results showing Snort alerts associated with suspicious activity from 172.31.0.50.

4.4 Investigation Findings and Limitations

According to the Infection Monkey report, the simulation discovered seven open services across four machines and successfully compromised one machine, TargetLinux02 (172.31.0.30), through a VSFTPD Backdoor exploit. The report also indicated that eight post-breach actions were performed on the compromised machine.

During my Splunk investigation, I identified several Snort alerts associated with possible Shellshock exploitation attempts originating from 172.31.0.50 and targeting 172.31.0.20 and 172.31.0.30. These alerts helped me identify suspicious activity and the machines involved, but they did not provide enough evidence to confirm that an attack was successful.

One important finding was that Splunk detected possible Shellshock attacks, while Infection Monkey reported a successful compromise through a VSFTPD Backdoor vulnerability. This highlighted a limitation of relying only on the available Snort alerts to determine the outcome of a security incident.

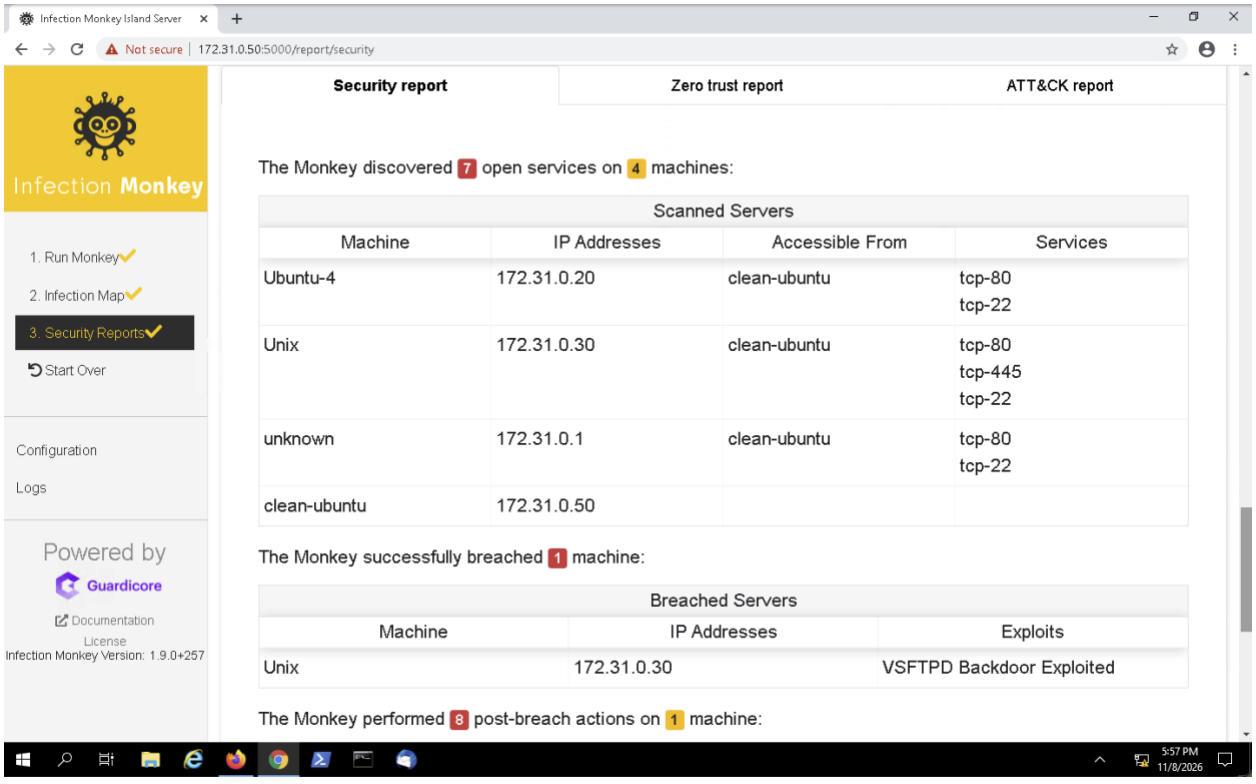


Figure *6. Infection Monkey security report showing the discovered services, successfully breached machine, and reported VSFTPD Backdoor exploitation.*

5. Security Recommendations:

Based on the findings from my Splunk investigation and the Infection Monkey security report, I identified several recommendations to improve network security and threat detection.

1. Patch and Update Vulnerable Services

The Infection Monkey report confirmed that TargetLinux02 (172.31.0.30) was successfully compromised through a VSFTPD backdoor vulnerability. I recommend removing or replacing vulnerable software, applying security updates, and disabling unnecessary services to reduce the risk of exploitation.

2. Improve Intrusion Detection and Prevention

Although Snort generated alerts related to possible Shellshock exploitation attempts, the available alerts did not confirm the successful VSFTPD backdoor exploitation. I recommend reviewing Snort detection rules, updating signatures, and evaluating whether IDS/IPS sensors have sufficient visibility into traffic between machines in the DMZ.

3. Improve Security Monitoring and Log Collection

I recommend collecting additional endpoint and authentication logs from the Linux machines and forwarding them to Splunk. This would help SOC analysts investigate suspicious processes, login attempts, and other system activities to determine whether an attack was successful.

These recommendations could help improve threat visibility, reduce security risks, and strengthen the organization's ability to detect and respond to future attacks.

6. Conclusion:

This project helped me understand how a SOC analyst uses Splunk and Snort to investigate suspicious network activity and identify potential security threats. During the investigation, I identified several alerts related to possible Shellshock exploitation attempts targeting multiple Linux machines. However, the Infection Monkey report confirmed that TargetLinux02 (172.31.0.30) was successfully compromised through a VSFTPD Backdoor exploit, which was different from the attacks identified in the Snort alerts.

One important lesson I learned is that detecting suspicious activity does not always mean an attack was successful. The investigation also showed the limitations of relying only on network security alerts to determine whether a system was compromised. Additional evidence from endpoint logs and other security tools may be necessary to confirm an incident.

Overall, this project improved my understanding of security monitoring, log analysis, incident investigation, and the importance of using multiple security tools to identify and respond to cybersecurity threats.